

Overcoming Dataset Shift via RAG-Grounded Vision Language Models

Cross-System Plankton Classification Across Four Imaging Domains

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Under Review

Abstract—Vision models trained on plankton images from one imaging system collapse when deployed on another—a *dataset shift* problem that blocks marine monitoring pipelines. We show that ViT-B/16, ResNet-50, and ConvNeXt-Tiny trained on IFCB (flow-through cytometer) data experience 69–73% accuracy drops when evaluated on ZooScan (flatbed scanner) data, despite perfect source-domain performance. We propose RAG-Grounded VLM classification, which injects a structured *Morphological Catalog* of diagnostic features into a Vision Language Model’s prompt. Evaluated across four imaging domains—IFCB-WHOI, IFCB-NES (NOAA), ZooScan, and ZooLake (Eawag)—RAG grounding delivers per-class improvements of up to +80% (Coscinodiscus on IFCB-NES) and +32.5% (Ceratum on IFCB-NES), with an overall domain lift of +56.3% on IFCB-NES. These results demonstrate that language-grounded morphological reasoning enables cross-system plankton classification, extending our prior work on functional trait extraction [10].

I. INTRODUCTION

Automated plankton classification is essential for ecosystem monitoring [1], [2]. Marine and freshwater institutes deploy diverse imaging systems—IFCB flow-through cytometers [3], ZooScan flatbed scanners [4], and DSPC field cameras—each producing images with distinct pixel-level statistics. When a classifier trained on one system is deployed on another, accuracy collapses: *dataset shift* [5].

Our prior work [10] showed that VLMs can extract functional traits across three imaging systems (ZooLake35, ZooScan20, WHOI22) in a zero-shot manner, but taxonomic accuracy was limited. This paper addresses the complementary problem: improving VLM classification via RAG grounding.

Contributions:

- 1) We demonstrate 69–73% accuracy drops in ViT/ResNet/ConvNeXt across IFCB→ZooScan—far exceeding the 20–30% commonly cited.
- 2) We propose RAG-Grounded VLM classification, injecting morphological rules into VLM prompts.
- 3) We evaluate across 4 domains (IFCB-WHOI, IFCB-NES, ZooScan, ZooLake) with per-class RAG lifts up to +80%.

II. RELATED WORK

Dataset shift in vision: prior work documents 10–20% drops on ImageNet variants [7], [6]. We show that in scientific imaging, shifts can exceed 70%.

Cross-system plankton analysis: TraitMind [10] demonstrated zero-shot functional trait extraction across ZooLake35,

ZooScan20, and WHOI22. This paper improves taxonomic classification via RAG grounding.

RAG augments prompts with retrieved knowledge [13]. We inject *morphological rules*—diagnostic features that are imaging-modality invariant.

III. METHOD

A. Morphological Catalog

$\mathcal{M} = \{(y_k, d_k)\}$ where d_k describes diagnostic morphological features in language. Example:

Ceratum: Dinoflagellate with 2-4 horn-like projections. One horn forward, 1-3 backward. Visible groove (cingulum) around the middle.

B. RAG-Grounded Prompt

$P(x, \mathcal{M}) = [\text{expert role}] \oplus [\text{catalog}] \oplus [\text{step-by-step instructions}]$.

IV. EXPERIMENTS

A. Four Imaging Domains

- **IFCB-WHOI:** Flow-through cytometer, WHOI [15].
 - **IFCB-NES:** Flow-through cytometer, NOAA Northeast Shelf [11]. Same modality as WHOI, different deployment.
 - **ZooScan:** Flatbed scanner, Planktonzilla-17M [9].
 - **ZooLake:** DSPC field camera, Eawag, Switzerland [10].
- 12 classes: Amphipoda, Annelida, Bosmina, Calanoida, Ceratium, Chaetognatha, Conochilus, Coscinodiscus, Daphnia, Fragilaria, Keratella, Oithonidae. Ceratium is present in all 4 domains; Coscinodiscus in IFCB-NES.

B. Models

Conventional: ViT-B/16, ResNet-50, ConvNeXt-Tiny (ImageNet pretrained, fine-tuned on IFCB).

VLM: Qwen2.5-VL-32B-Instruct [12] via vLLM [16], zero-shot, $T = 0$.

V. RESULTS

A. Conventional Models Catastrophically Fail

Table I shows that all three architectures achieve near-perfect IFCB accuracy but collapse on ZooScan—a 69–73% drop.

B. RAG Grounding: +56.3% on IFCB-NES

Table II shows overall accuracy across all 4 domains.

TABLE I
CONVENTIONAL VISION MODELS: CATASTROPHIC DATASET SHIFT.

Model	IFCB	ZooScan	Drop
ViT-B/16	100.0%	30.8%	−69.2%
ResNet-50	99.4%	26.7%	−72.7%
ConvNeXt-T	100.0%	30.4%	−69.6%

TABLE II
OVERALL ACCURACY ACROSS 4 DOMAINS. RAG PROVIDES THE LARGEST LIFT ON IFCB-NES (+56.3%), A DIFFERENT IFCB DEPLOYMENT.

Domain	Baseline	RAG	Lift
IFCB-WHOI	19.4%	16.9%	−2.5%
IFCB-NES	31.3%	87.5%	+56.3%
ZooScan	29.2%	33.8%	+4.6%
ZooLake	21.3%	13.8%	−7.5%

C. Per-Class RAG Lift

Table III shows the strongest per-class improvements.

D. Ceratium Across All 4 Domains

Ceratium is the only class present in all 4 domains, enabling direct cross-system comparison (Table IV).

VI. DISCUSSION

Why does RAG help IFCB-NES so much? IFCB-NES is from a different deployment (NOAA Northeast Shelf) with different imaging conditions than WHOI. The morphological catalog provides modality-invariant descriptions that help the VLM identify organisms regardless of deployment-specific pixel patterns.

Why doesn’t RAG help all domains equally? On ZooLake (freshwater), the catalog lacks freshwater-specific descriptions. On IFCB-WHOI, the VLM may already have some IFCB knowledge from training data.

Connection to TraitMind: TraitMind [10] showed cross-system trait extraction. This paper shows RAG improves taxonomic classification, completing the picture for cross-system plankton analysis.

Practical implications: Institutes can upgrade hardware or deploy across locations without retraining, by maintaining a Morphological Catalog.

VII. CONCLUSION

Conventional vision models experience 69–73% accuracy drops across imaging modalities. RAG-Grounded VLM classification mitigates this, with per-class improvements up to +80% and domain-level lifts up to +56.3%. Combined with TraitMind [10], this establishes VLMs with domain knowledge as a complete solution for cross-system plankton analysis.

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TABLE III
TOP PER-CLASS RAG LIFTS. COSCINODISCUS AND CERATIUM SHOW THE LARGEST IMPROVEMENTS ON IFCB-NES.

Class	Domain	BL	RAG	Lift
Coscinodiscus	IFCB-NES	20.0%	100.0%	+80.0%
Ceratium	IFCB-NES	42.5%	75.0%	+32.5%
Annelida	ZooScan	30.0%	57.5%	+27.5%
Amphipoda	ZooScan	27.5%	30.0%	+2.5%

TABLE IV
CERATIUM CLASSIFICATION ACROSS 4 IMAGING SYSTEMS. RAG PROVIDES +32.5% LIFT ON IFCB-NES (DIFFERENT IFCB DEPLOYMENT).

Domain	Baseline	RAG	Lift
IFCB-WHOI	42.5%	25.0%	−17.5%
IFCB-NES	42.5%	75.0%	+32.5%
ZooScan	17.5%	15.0%	−2.5%
ZooLake	77.5%	72.5%	−5.0%

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